

Logan Shaffer

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PROFESSIONAL SUMMARY

A recent Engineering Physics graduate from Embry-Riddle Aeronautical University with extensive experience in full-stack simulation development, real-time data acquisition, and extended reality (XR) development. Experienced in the software development life-cycle, asynchronous data flow, and collaboration in Agile environments. Seeking to apply problem-solving and software design skills to a full-time software development position.

CORE SKILLS

Programming Languages: Proficient in C#, Python, HTML, CSS, JavaScript, React. Intermediate in C, C++

Software & Simulation: Unity 6, Blender, CATIA, WebGL, Three.js, Cesium, Node.js, REST APIs, Git/GitHub

EDUCATION

Embry-Riddle Aeronautical University, Daytona Beach Campus

Bachelor of Science (B.S.) in Engineering Physics

Spacecraft Instrumentation Track and minors in **Electrical & Computer Engineering** and **Applied Mathematics**

EXPERIENCE

NASA Ames Research Center

Moffet Field, CA

OSTEM Intern: Simulation Developer

June 2026 – Present

- Improved underlying logic of the simulator through modularization of the **JavaScript** backend as well as patching known bugs and issues, resulting in an improved experience for consistent government users such as **Civil Air Patrol** and Educators.
- Modernized the frontend of the simulator web version and cross platform mobile app using **React** and **Tailwind CSS**, resulting in an improved look and feel of the program.
- Incorporated a research grade simulation tool in development at Ames as well as collaborated with an FAA ATC specialist.

Embry-Riddle Aeronautical University

Daytona Beach, FL

Extended Reality (XR) Developer

May 2025 – May 2026

- Developed the “Immersive Flight Dynamics” program in **Unity**, implementing a **UDP server** to facilitate communication between the application, external **Python** scripts, and a **YAW Motion Simulator**.
- Integrated **JavaScript** for asynchronous file I/O operations and processed live ATSB data for real-time simulation accuracy.
- Collaborated with faculty and experienced professionals to iterate on **UX design**, improving simulation immersion and developing robust back-end safety protocols for edge-case scenarios.
- Incorporated **Agile** methodologies to manage development, utilizing **Git** for ensuring maintainable, high-quality code.

Research Experience for Undergraduates (REU) Intern, “A Versatile Synthesis of Self-Healing Polymers” May 2024 – Jul 2024

- Developed **Python** scripts to automate force and distance data analysis, reducing weekly processing time by **10+ hours**.
- Conducted material characterization of self-healing PDMS polymers through tensile testing and experimental refinement.

PROJECTS

Senior Design Capstone, mDLP (miniature Double Langmuir Probe):

- Engineered a real-time scientific data acquisition system using **Python** and **MVC Pattern** to process scientific measurements intended for ionospheric research and plasma physics.
- Implemented asynchronous data flow to handle direct serial communications and high-speed simulation fallbacks.
- Designed a threaded process manager to coordinate data sources, recording, and playback, ensuring high performance.

Junior Design Project, Ball Balancing Platform

Jan 2025 – Apr 2025

- Programmed inverse kinematics and **PID control algorithms** in **C++** using **Arduino** to achieve precise autonomous platform control capable of balancing a steel ball atop itself.

Microcomputers & Electronics Final Project, Project S.H.I.N.E

Jan 2025 – Apr 2025

- Created a CubeSat adjacent light tracker using **C** and an **AVR micro controller**, implementing real-time telemetry to monitor power, thermal, and orientation data via a remote monitoring station.

ADDITIONAL

Honors: Embry-Riddle Honors Program, Tau Beta Pi Engineering Honor Society (*Corresponding Secretary*), Omicron Delta Kappa National Honor Society, *Vice President*, Embry-Riddle Presidential Scholarship, College of Arts and Science Dean’s List

Leadership: Student Union Advisory Board, *Point of Contact*